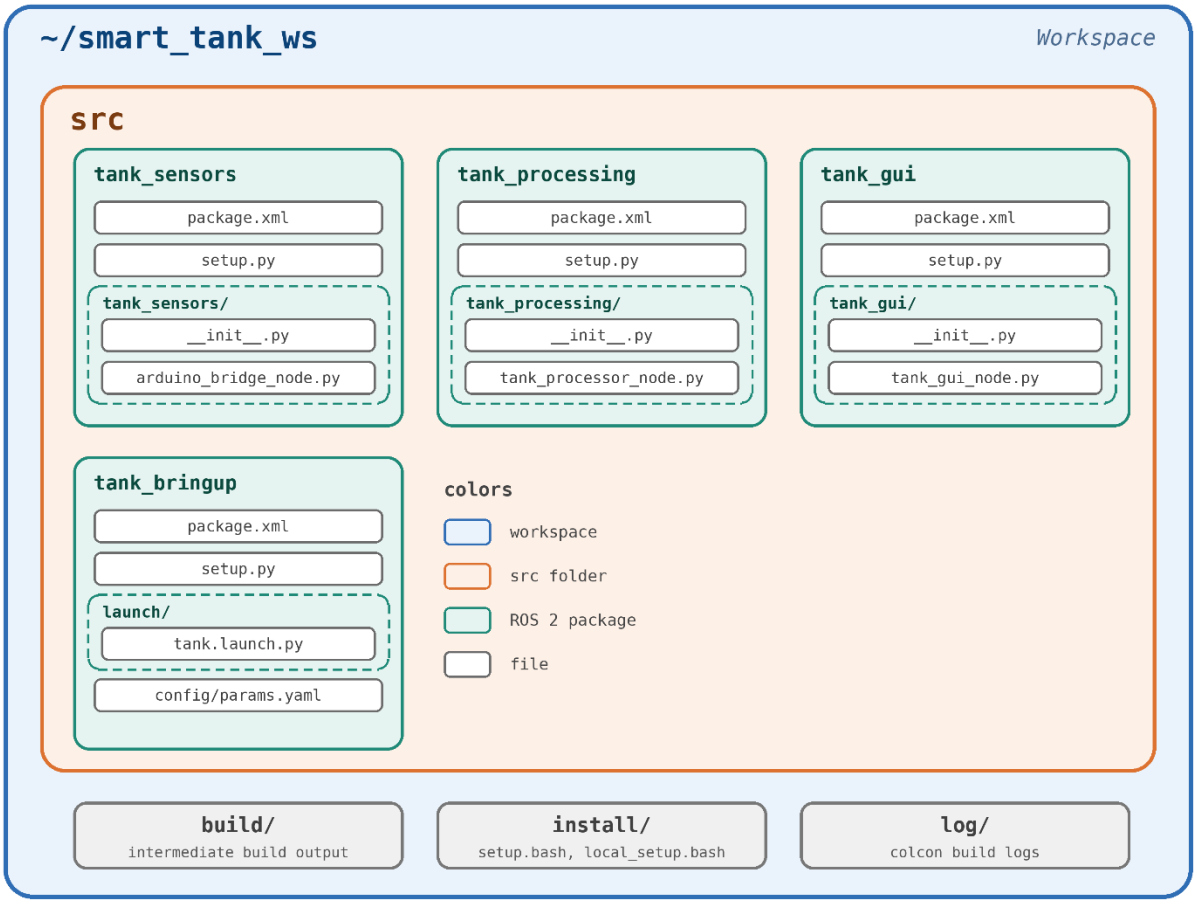
ROS 2 Smart Tank System

Workspace, Nodes, and Topics

1. ROS 2 Workspace Structure

We will organize the project into a ROS 2 workspace named smart_tank_ws, located at ~/smart_tank_ws.



The workspace will contain four ROS 2 packages inside the src directory:

- **tank_sensors**: We will use this package to implement arduino_bridge_node.py, which will communicate with the Arduino or ESP32 and receive sensor measurements.
- **tank_processing**: We will use this package to implement tank_processor_node.py, which will process sensor measurements and calculate the required tank parameters.
- **tank_gui**: We will use this package to implement tank_gui_node.py, which will display sensor readings and calculated results through a graphical interface.
- **tank_bringup**: We will use this package to organize the launch configuration that will allow us to start the system's nodes together.

We will use colcon to build the workspace. This process will generate the build, install, and log directories.

2. ROS 2 Nodes

A node will be an independent program that performs a specific task within our ROS 2 system. We will implement three main functional nodes.

2.1 Arduino Bridge Node

- **Package:** tank_sensors
- **File:** arduino_bridge_node.py
- **Function:** We will use this node to communicate with the Arduino or ESP32, receive weight, ultrasonic distance, and temperature measurements, and publish the readings to ROS 2 topics.

2.2 Tank Processor Node

- **Package:** tank_processing
- **File:** tank_processor_node.py
- **Function:** We will use this node to receive sensor measurements, validate the data, calculate liquid volume and fill percentage, and publish the processed results.

2.3 Tank GUI Node

- **Package:** tank_gui
- **File:** tank_gui_node.py
- **Function:** We will use this node to display sensor measurements, calculated tank parameters, and system status through a graphical interface.

3. ROS 2 Topics

A topic will be a named communication channel through which ROS 2 nodes will exchange messages. Our nodes will publish messages to topics or subscribe to topics to receive data.

We will use the following proposed topics for the Smart Tank System.

3.1 Sensor Topics

- **/tank/sensors/weight_kg** — We will use this topic to publish weight measurements in kilograms.
- **/tank/sensors/temperature_c** — We will use this topic to publish temperature measurements in degrees Celsius.
- **/tank/sensors/distance_m** — We will use this topic to publish ultrasonic distance measurements in metres.

The Arduino Bridge Node will publish these three topics, and the Tank Processor Node will subscribe to them to receive the sensor measurements.

3.2 Processing Topics

- **/tank/volume_l** — We will use this topic to publish the calculated liquid volume in litres.
- **/tank/fill_percentage** — We will use this topic to publish the estimated tank fill percentage.
- **/tank/status** — We will use this topic to publish the current tank status and relevant warnings.

The Tank Processor Node will publish these topics, and the Tank GUI Node will subscribe to them to display the calculated information.

The Tank GUI Node will also subscribe directly to the sensor topics so that we can display the raw measurements alongside the calculated values.